

1. Objectives

To study how an investment subsidy affects firm behaviour in an economy where firms face costs when adjusting their capital stock. The goal is to build a model, calibrate it to match real data on investment and compare the economy with and without 10% subsidy.

2. The Model

Firm Problem

- Each firm maximizes the discounted sum of profits

$$\max_{\{i_{j,t}, h_{j,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \pi_{j,t},$$

- Output is Cobb-Douglas with decreasing returns
- Productivity z follows an AR(1) in logs:

$$\ln z_{j,t+1} = \rho \ln z_{j,t} + (1 - \rho) \ln \bar{z} + \epsilon_{j,t+1}, \quad \epsilon_{j,t+1} \sim \mathcal{N}(0, \sigma^2),$$

Optimal labor (closed form)

We solve for optimal labor from the static FOC: $h^* = \left(\frac{\nu z k^\alpha}{w} \right)^{\frac{1}{1-\nu}}$

Plugging h^* back to the profit function gives operating profit as a function of k and z only, which simplifies the problem a lot.

Bellman equation

The firm's problem in recursive form is:

$$V(k, z) = \max_{k'} \{ \tilde{\pi}(k, z) - C(k, k') + \beta \mathbb{E}[V(k', z') | z] \}$$

where $\pi_{\text{tilde}}(k, z)$ is the reduced-form profit after substituting h^* and $C(k, k')$ is the total investment cost.

3. How do we solve?

Grids

- Capital: 200 points, log-spaced between 0.01 and 50. Log-spacing will put more points near zero, where the value function curves the most.
- Productivity: 9 points using the Rouwenhorst method. The grid is shifted, so that $\mathbb{E}[z] = 1$

Value Function Iteration (VFI)

We pre-compute the full payoff tensor once and then iterate:

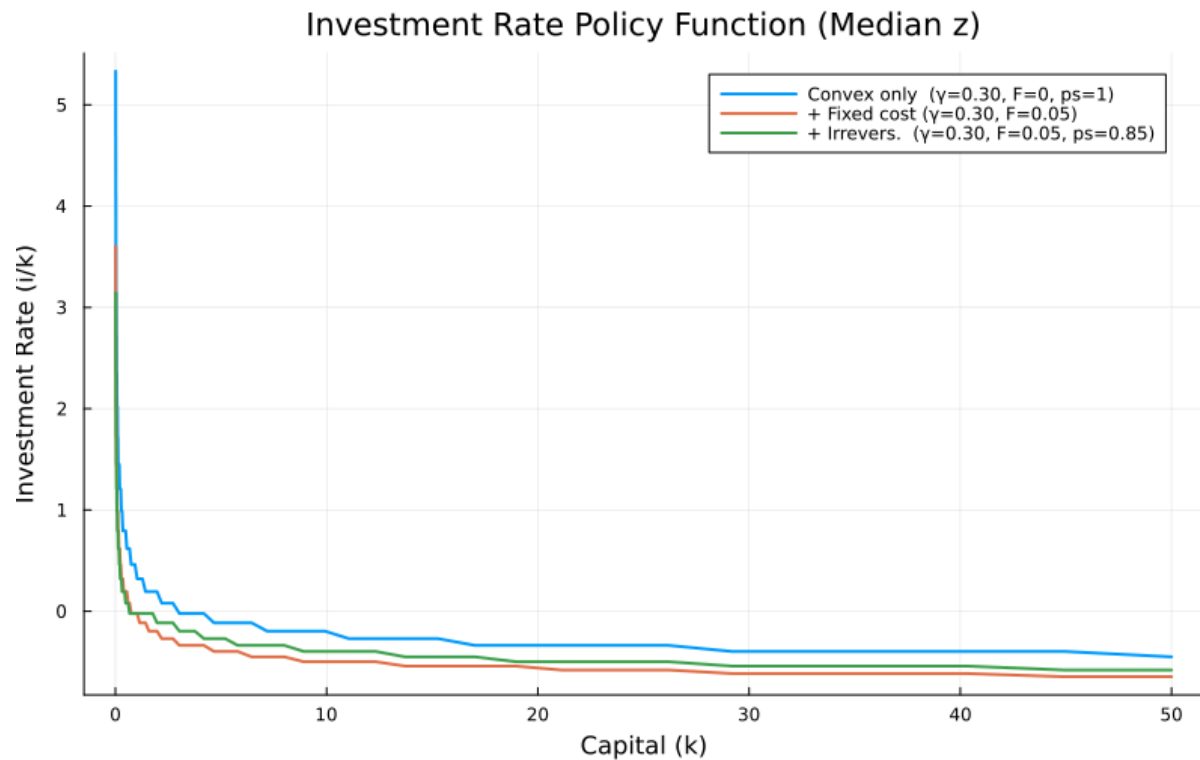
- Compute expected continuation: $EV = V * P'$
- For each z we find max over k' of: $R[k, k', z] + \beta EV[k', z]$
- We repeat until $\max |V_{\text{new}} - V_{\text{old}}| < 1e-6$

We do not use the endogenous grid method, because of the fixed costs.

Simulation

After solving for the policy function, we simulate 5000 firms for 600 periods and delete the first 100 as burn-in.

4. Exploration



All 3 curves start very high for small capital and decline steeply, then become flatter at slightly negative values. This makes sense as small firms have a high MPK so they invest more, large firms are around their target and let the capital depreciate.

- At high k : the blue curve (convex only) is most negatively sloped. The orange curve adding fixed costs makes it slightly flatter. Green curve, which adds irreversibility makes it the flattest, firms are less willing to sell capital, as they only get 0.85 back.
- At low k : the curves are almost the same, which makes sense. When firms are undercapitalised they will invest without looking at the cost structure.

5. Calibration

Fixed parameters

Parameter	Value
alpha	0.30
nu	0.60

delta	0.08
r	0.04
w	1.00
rho	0.90
sigma	0.12

Estimated Parameters (SMM)

We estimate gamma(convex cost), F(fixed cost), and ps(resale price) by matching moments to data. Estimator minimizes:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \left[m^{\text{model}}(\theta) - m^{\text{data}} \right]' W \left[m^{\text{model}}(\theta) - m^{\text{data}} \right]$$

Calibrated values are:

- gamma = 0.35
- F = 0.05
- ps = 0.85

Moment fit

Moment	Data	Model	Difference
Avg Investment Rate	12.2%	11.2%	-8.2%
Inaction Rate	8.1%	28.9%	256.3%
Fraction Negative Inv.	10.4%	49.7%	378.2%
Positive Spike	18.0%	23.7%	31.9%
Negative Spike	1.4%	7.0%	400.6%

SMM optimizer converged to a bad solution. We also tried the benchmark estimates from the quoted Cooper and Haltiwanger (2006) paper, but it gave us even worse results. Therefore we decided to use gamma=0.35, F=0.05, ps=0.85 as this produces the closest fit.

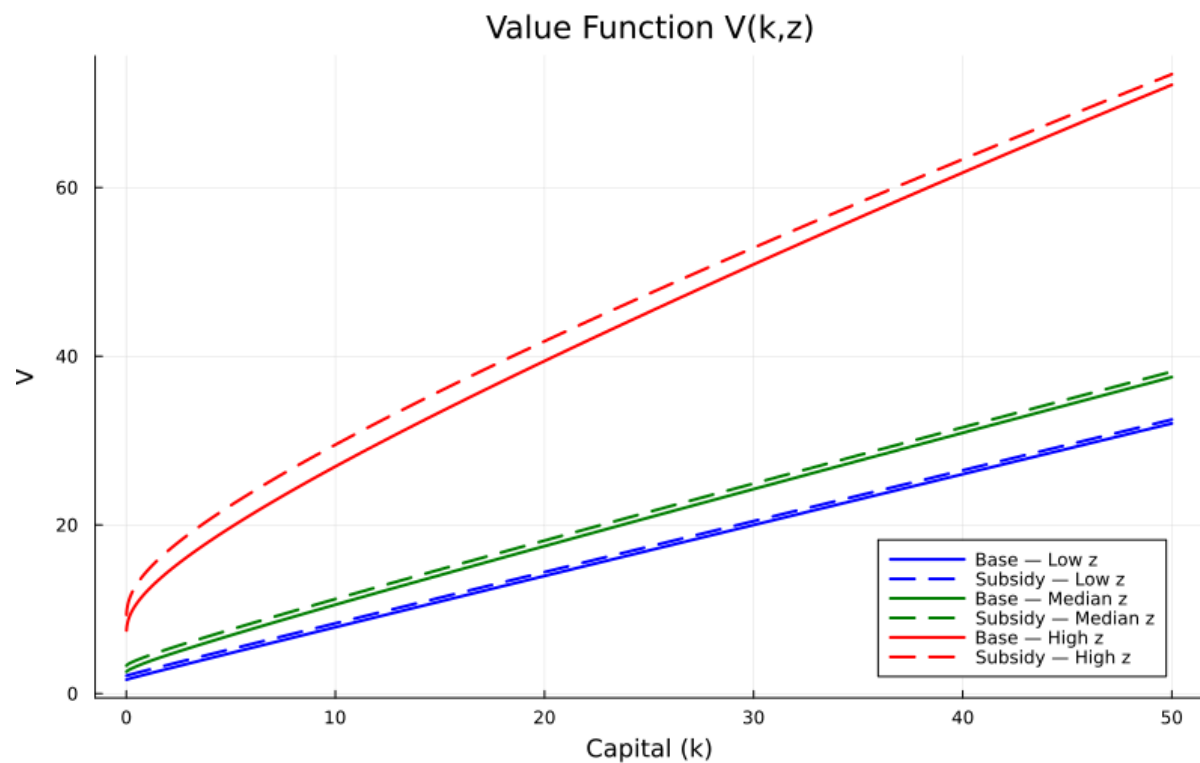
Grid Sensitivity

Nk	Nz	AvgInv	Inaction	PosSpike	NegSpike
60	5	0.1131	0.0	0.2873	0.0465
120	7	0.1128	0.0	0.2485	0.0714
200	9	0.1126	0.2891	0.2372	0.0696

Key moments are stable across grid configurations, the inaction rate is the only exception, therefore we use Nk=200, Nz=9 as the best grid for all reported results.

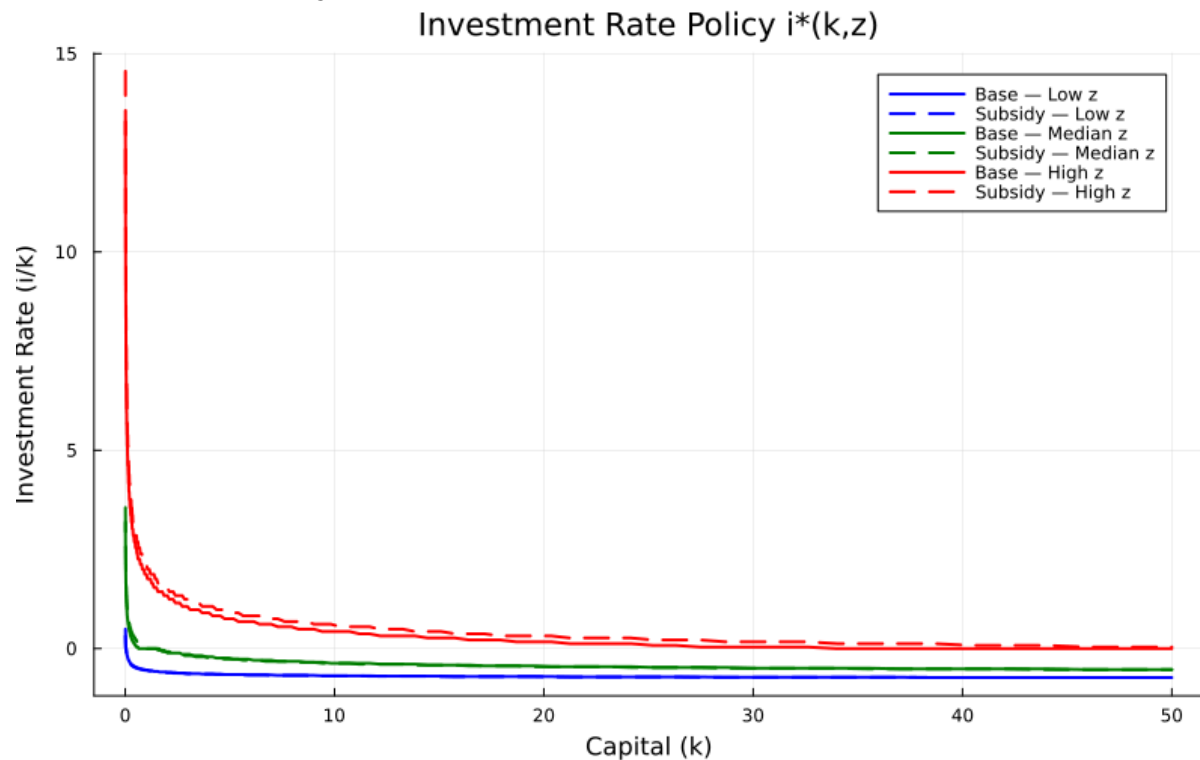
6. Policy Functions

Value Function



The Value Function $V(k,z)$ is increasing and concave in capital for all productivity levels. More capital is always better, but with diminishing returns. Higher productivity, z , shifts the entire function up as more productive firms earn higher profits for given capital. Dashed lines representing subsidy raises firm value across all pairs. The effect is largest for red curve high-productivity firms, because firms invest more and therefore benefit more from the lower price of capital.

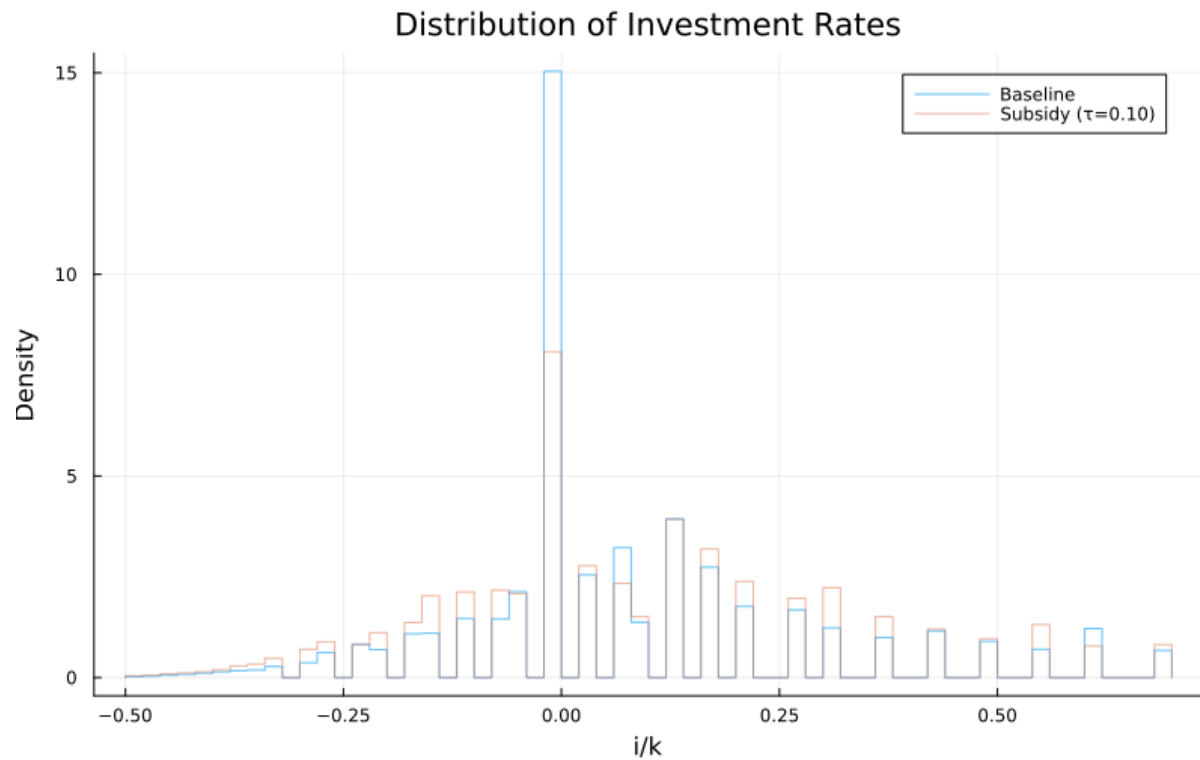
Investment Rate Policy



The Investment Rate Policy function shows firms investing heavily at low capital stocks, with the rate declining steep as capital increases. High productivity firms (red) invest significantly more than low productivity firms (blue) at every capital level, because they have higher target capital stock. The subsidy shifts investment rates upwards for all capital and productivity. The effect, again, is most visible for high productivity firms, and the most significant at the middle part.

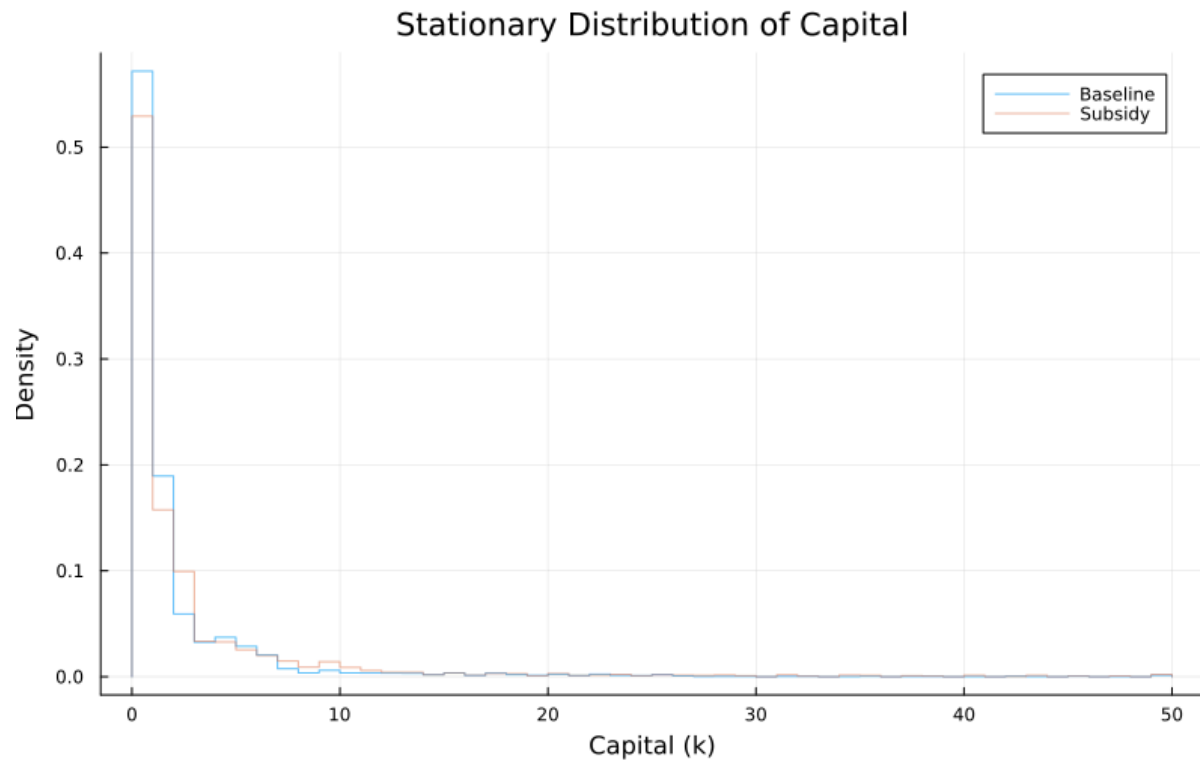
7. Stationary Distribution

Investment Rates



The investment rate distribution shows a large point at $i/k=0$, confirming the inaction region generated by fixed cost. The subsidy (orange) reduces the spike significantly, as firms are pulled out of inaction by lower prices. The subsidy also shifts mass to positive investment rates with more firms making positive investments.

Capital



Capital distribution is heavily right skewed, meaning that most firms hold small capital stocks. After the subsidy, the two distributions look similar, but the data shows that subsidy raises aggregate capital by 31.67%.

8. Policy Analysis 10% subsidy

Investment Moments

Moment	Data	Baseline	Subsidy
Avg Investment Rate	12.2%	11.2%	11.9%
Inaction Rate	8.1%	28.9%	15.4%
Negative Inv	10.4%	49.7%	44.2%
Positive Spike	18.0%	23.7%	29.6%
Negative Spike	1.4%	7.0%	10.2%

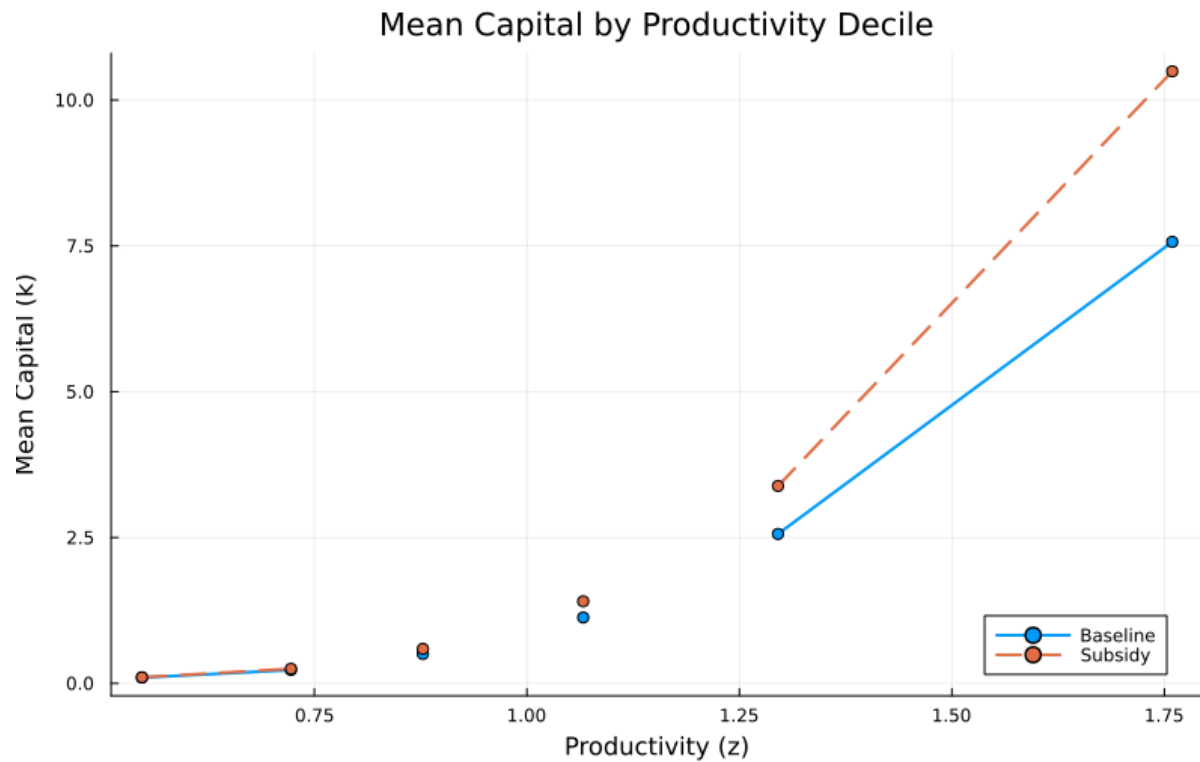
The subsidy raises the average investment rate and reduces inaction as firms are pulled out of inaction by the lower purchase price. The positive Spike rate raises as more firms make large investments.

Aggregates

Variable	Baseline	Subsidy	% Change
Output Y	1.3898	1.7212	+23.84%
Capital K	2.1958	2.8911	+31.67%
Labor H	0.8339	1.0327	+23.84%
Subsidy Cost / Y	-	2.30%	-

The subsidy raises aggregate capital by 31.67% and output by 23.84%. Labor also rises as higher capital increases MPL. Subsidy cost is 2.30% of output, showing 10x multiplier.

Cross Sectional Patterns



The plot confirms that more productive firms hold more capital in both economies. The subsidy raises capital holdings across all productivity levels, but the effect is strongest for high productivity ones. This is consistent with the rise in $\text{Corr}(k,z)$ from 0.592 to 0.621, meaning the subsidy improves efficiency by allocating capital towards most productive firms.